Lab 3 – Building an Interactive Dashboard in Tableau

In this lab, you'll learn how to build an **interactive dashboard** in Tableau to support **wine recommendation and exploration**, targeted at new wine consumers ("wine beginners"). You'll combine multiple charts, filters, and actions to build a tool that allows users to explore wines by category, region, rating, and price.

By the end of this lab, you will:

- Understand what makes a dashboard effective
- Create charts that support interaction and user-driven filtering
- Use calculated fields to group data (e.g., rating classifications)
- Build a clean, purpose-driven dashboard focused on wine selection. Combine multiple charts and filters for intuitive exploration.

> **Main goal**: Help users answer the question **"What wine should I buy?"**, based on their preferences (e.g., wine type, quality, price, region).

We continue using the same datasets from Lab 1: `winemag-data-with-year.csv` and `wine_category.csv`.

What Makes a Good Dashboard?

Before you start, here's a quick checklist for **effective dashboards**:

- **Clear Purpose** – Focused on a specific question or goal. Focused metrics aligned with the user's goals
- **Tells a Story** – Charts work together to reveal insights
- **Logical Layout** – Clear visual hierarchy, clean formatting
- **Consistent Design** – Fonts, color schemes, and styles align
- **Useful Interactions** – Filters and highlights enhance the experience

Common Pitfalls:

- Too many charts or texts with no visual focus
- Inconsistent font or color use
- Irrelevant or redundant charts that don't help answer the main question
- Poor design: pie charts with many slices, no axis labels, misleading visuals

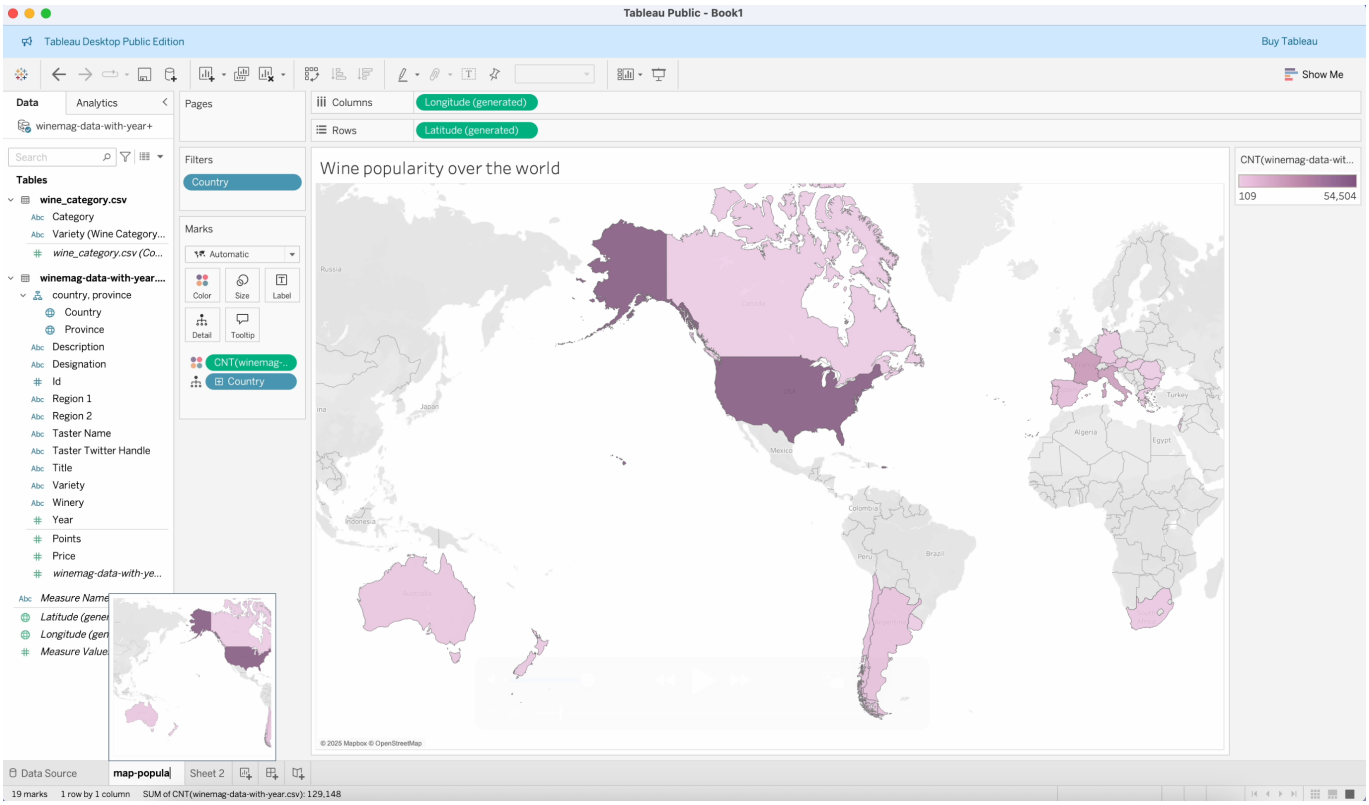
Build the Wine Selector Tool

1. Import Data

- Open Tableau and import both datasets and **join on** `variety`
- You may reuse your workbook from previous labs

2. Create Related Visualizations

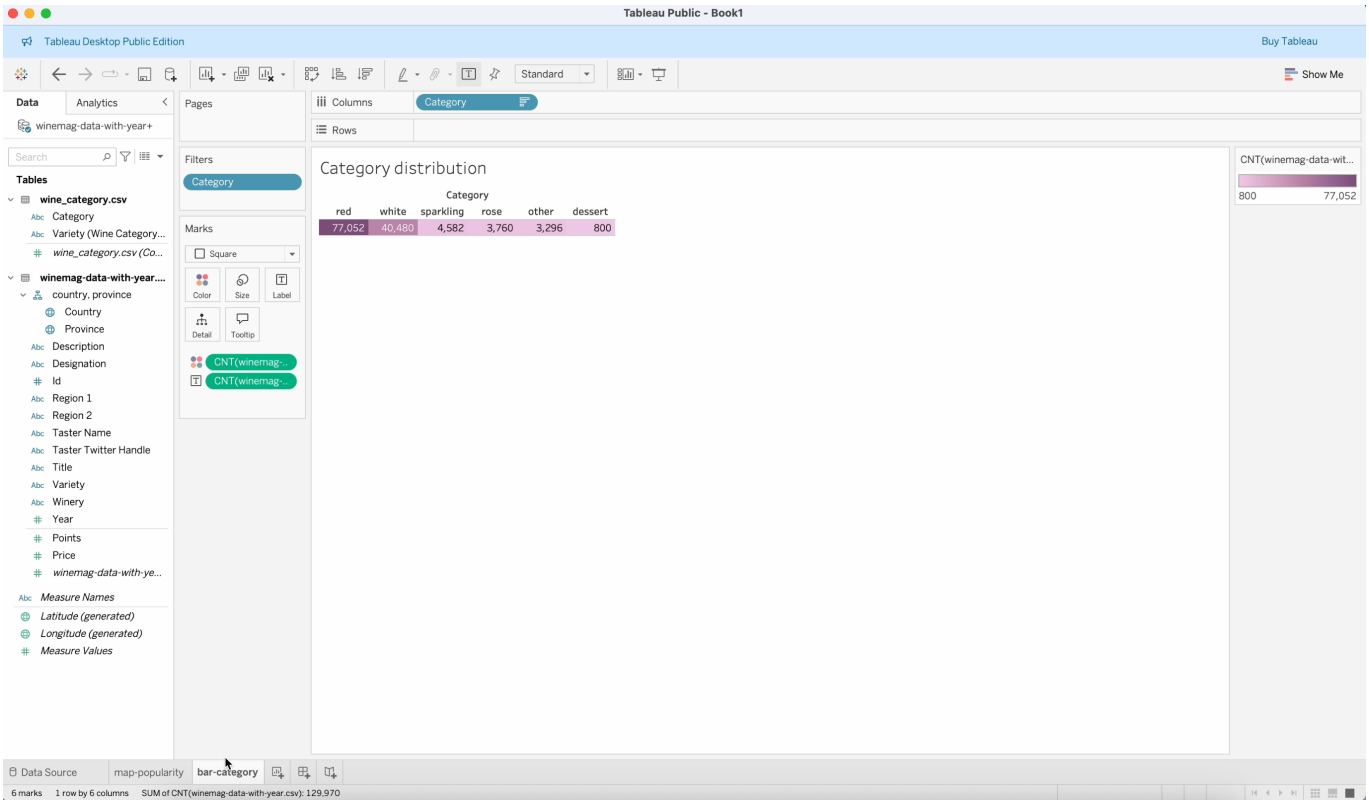
2.1 Map – Wine Review Distribution by Country



Create a map showing wine **popularity** using number of reviews.

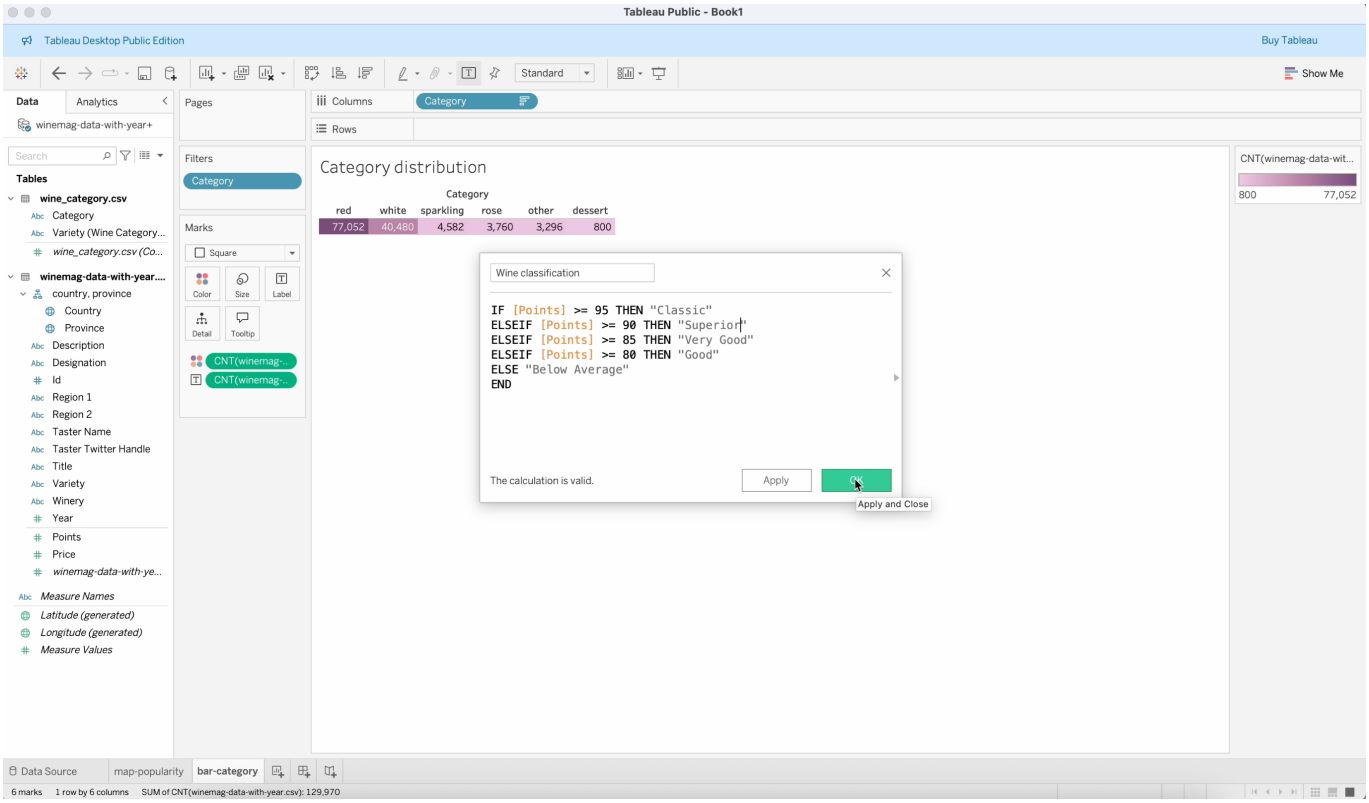
- Use `country` → automatic map
- Use `number of reviews` for color
- Filter to show only countries with **>100 reviews**
- Rename the sheet to **Map: Wine Popularity by Country** so it's clear when adding it to your dashboard.
- Customize color palette and map style

2.2 Highlighted Table – Wine Category Distribution (Global)



- Highlighted Table of `category` vs `number of reviews`
- Customize colors for red/white/sparkling/etc (e.g., based on `number of reviews`)
- This helps users understand which types are most common

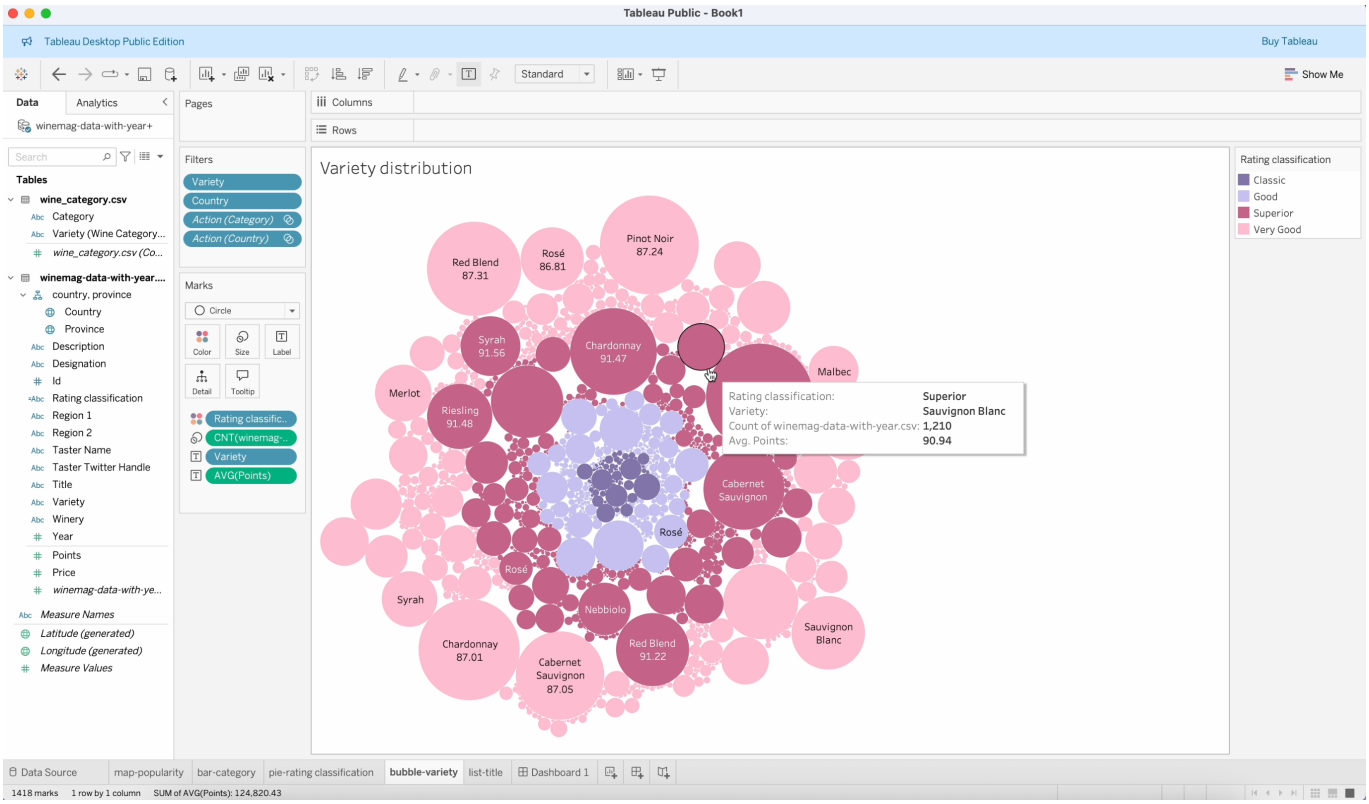
2.3 Create a Calculated Field – Rating Classification



Help users interpret wine ratings more easily by grouping `points`.

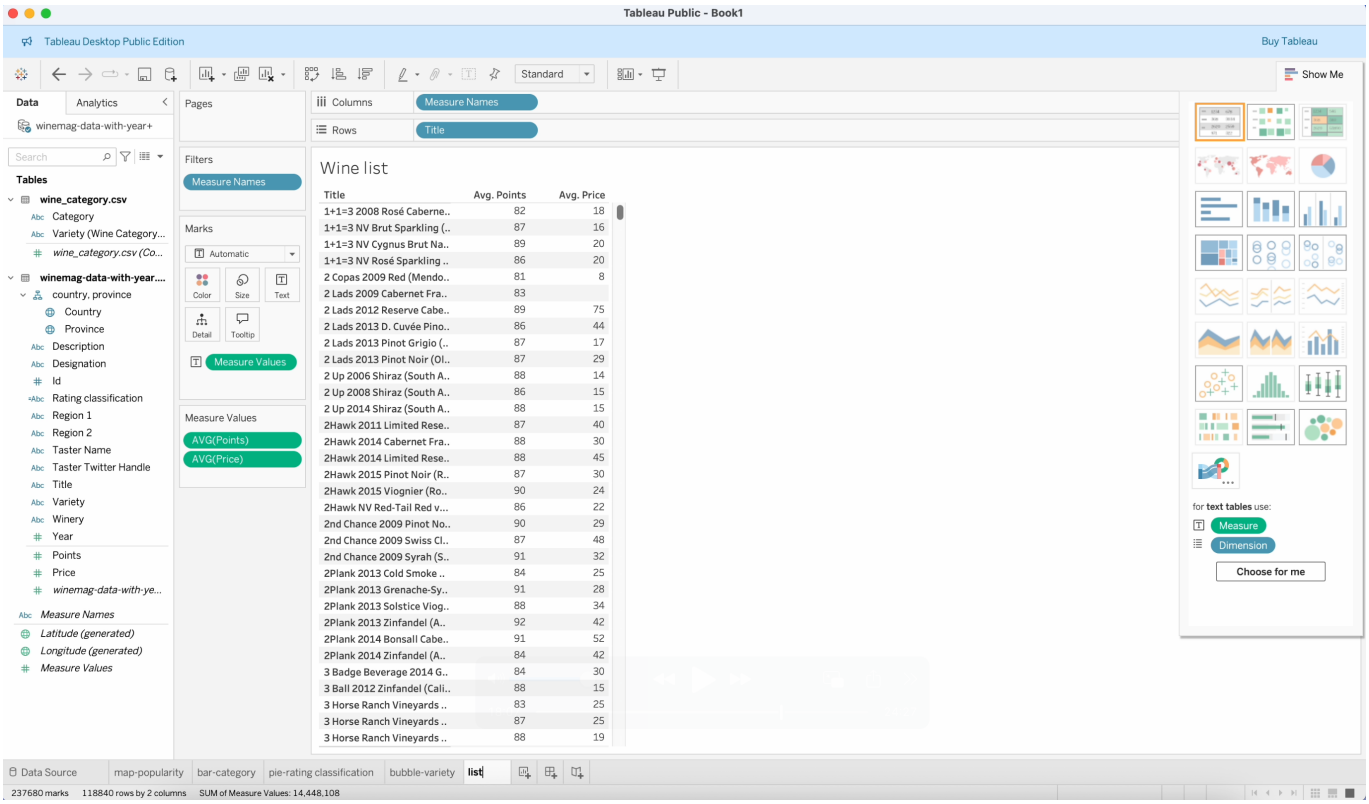
- Go to *Analysis* > *Create Calculated Field*
- Categorize points to `classic`, `superior`, `very good`, `good`.
  You can now use this field for filtering or color coding.

2.4 Bubble Chart – Wine Variety (Multi-dimensional)



- Size: number of reviews
- Color: **Rating Classification**
- Labels: `variety` + average rating
- Use this to help users identify popular top-performing wine types

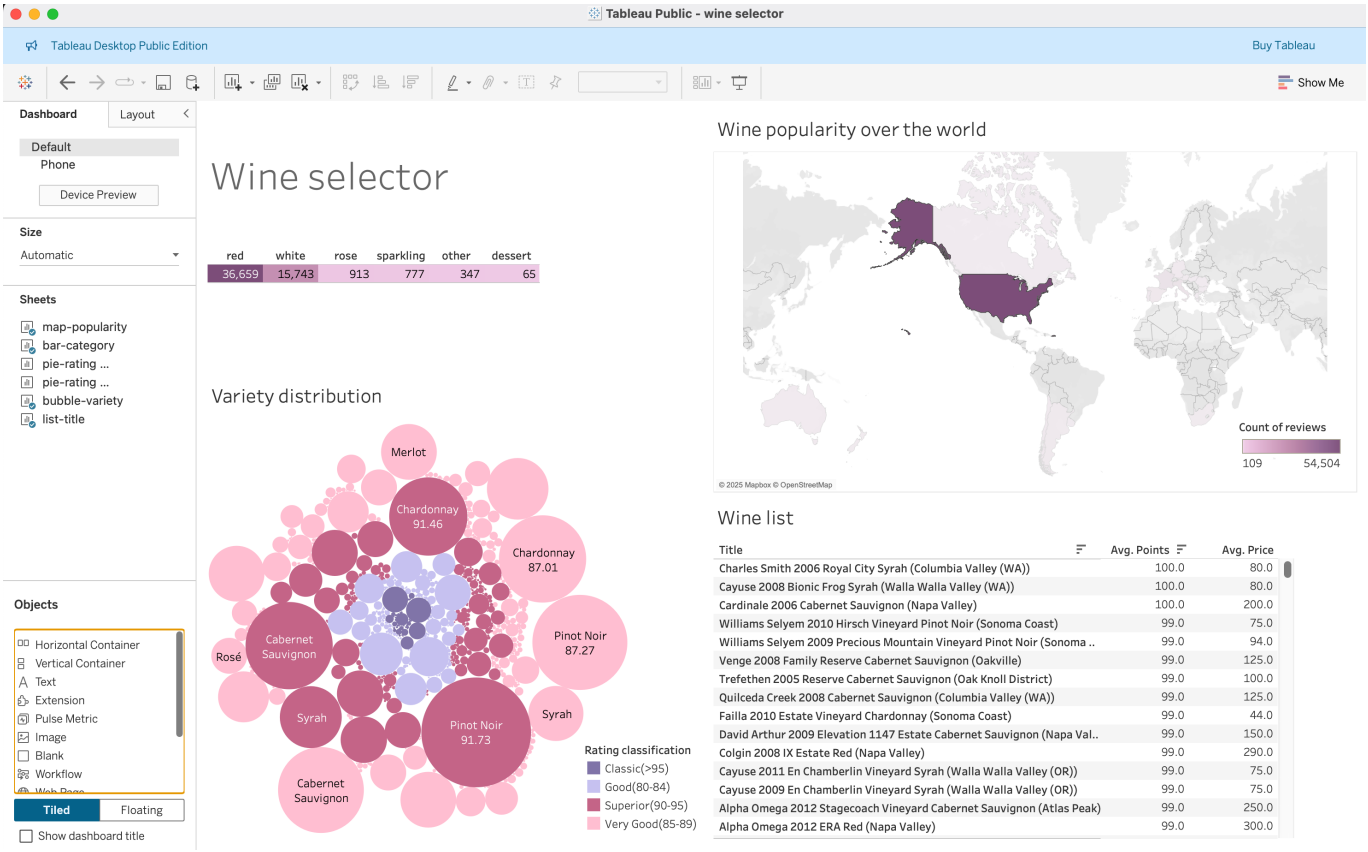
2.5 Wine Detail Table – Title, Price, Rating



- Create a table of `title`, `price`, `points`
- Sort by price or rating to help users **make choices**

3. Assemble Everything and Make It Interactive

3.1 Add Charts to a Dashboard



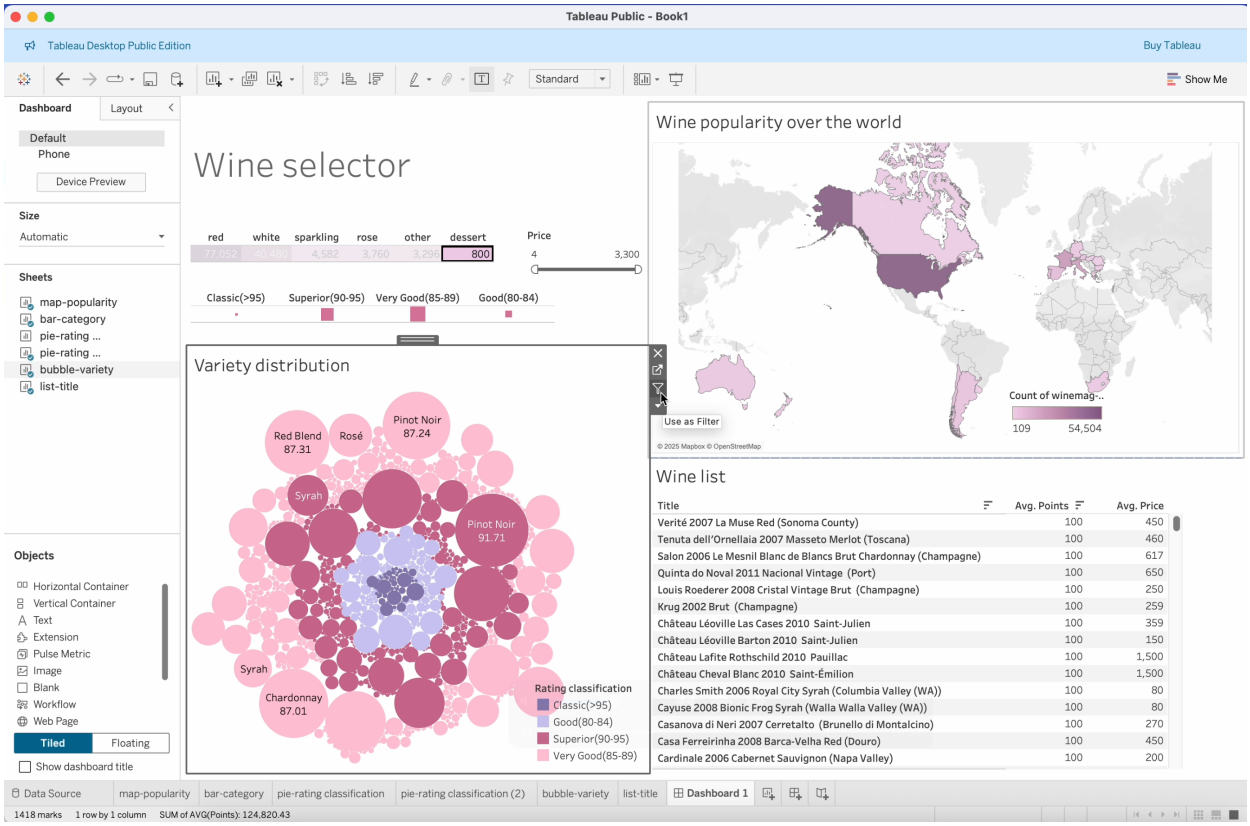
- Create a new **Dashboard** and drag in your 4 key visualizations:
  1. Map (Country Popularity)
  2. Highlighted table (Category)
  3. Bubble (Variety)
  4. List (Wine title/price/rating)
- Add a clear **title** (e.g., "Wine Selector")
- Use containers to keep layout clean and organized
- You may:
  - Set legends as **floating** to save space
  - Hide sheet titles if already obvious from context
  - Add **captions/annotations** for key insights

3.2 Add Filters for Interactivity

Use filters to focus the data displayed according to user preferences. Filters help users narrow down their selection by excluding data that does not meet specific criteria, making it easier to find wines that match their desired criteria.

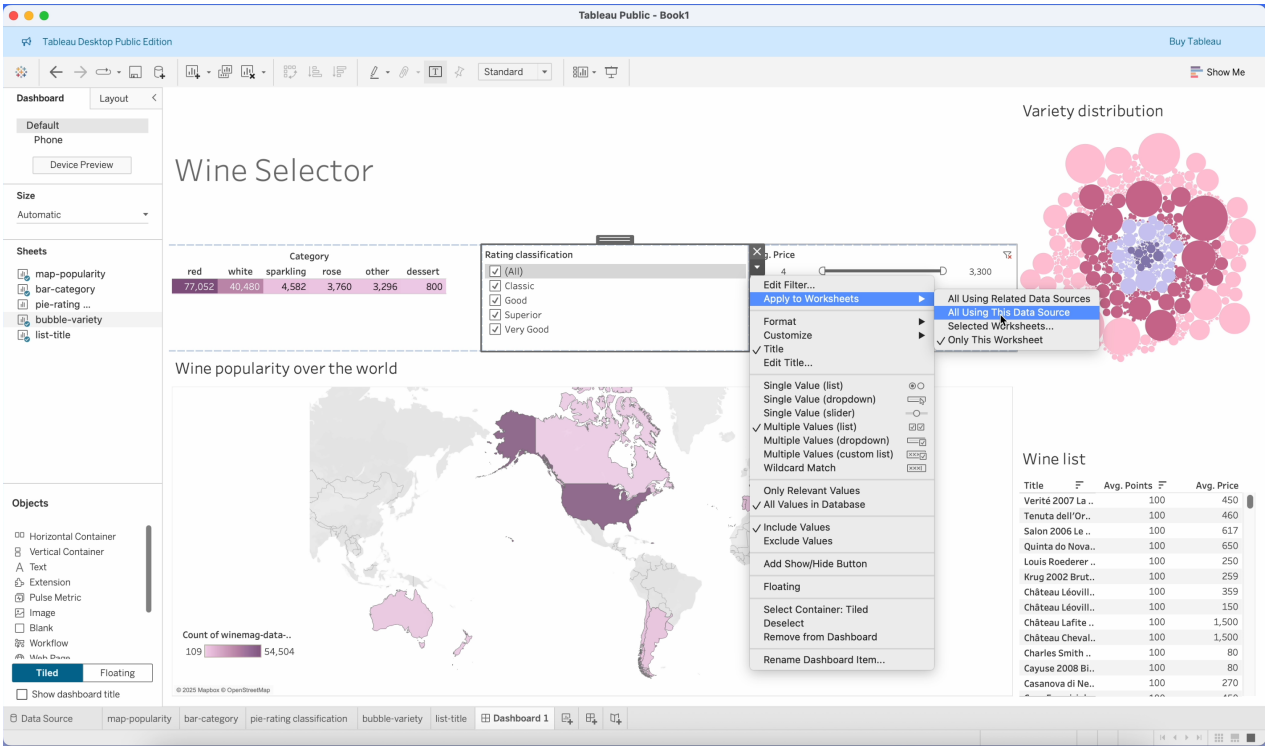
There are two types of filters to set:

- For fields already shown in charts (e.g., country, category and variety):
  - On the chart → Click the side menu → **Use as Filter**
  - Example: Clicking a country on the map filters all other charts



- For attributes not shown directly in the chart (price, rating classification):
  1. Go to one chart that contains the attribute (e.g., Wine List). Note that you must first add that attribute to the chart as Detail or Filter before it becomes available.
  2. Open the dropdown → **Show Filter**. The filter will be visible.

3. On the filter dropdown → Choose **“Apply to Worksheets” → “All Using This Data Source”**

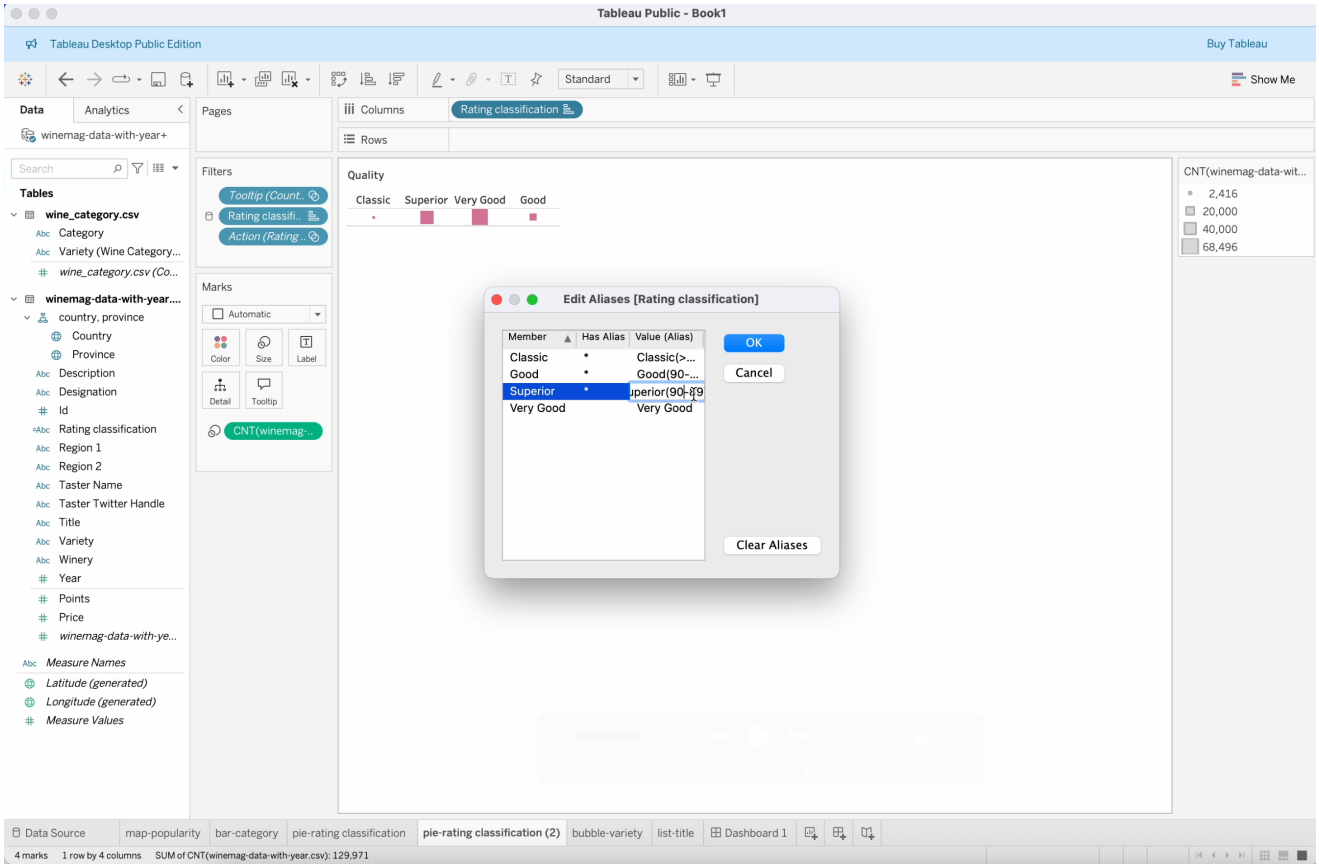


Tip: Horizontal Filter UI

Tableau's multi-select filters are vertical by default. To create a **horizontal filter interface**, use a horizontal **table** as a workaround.

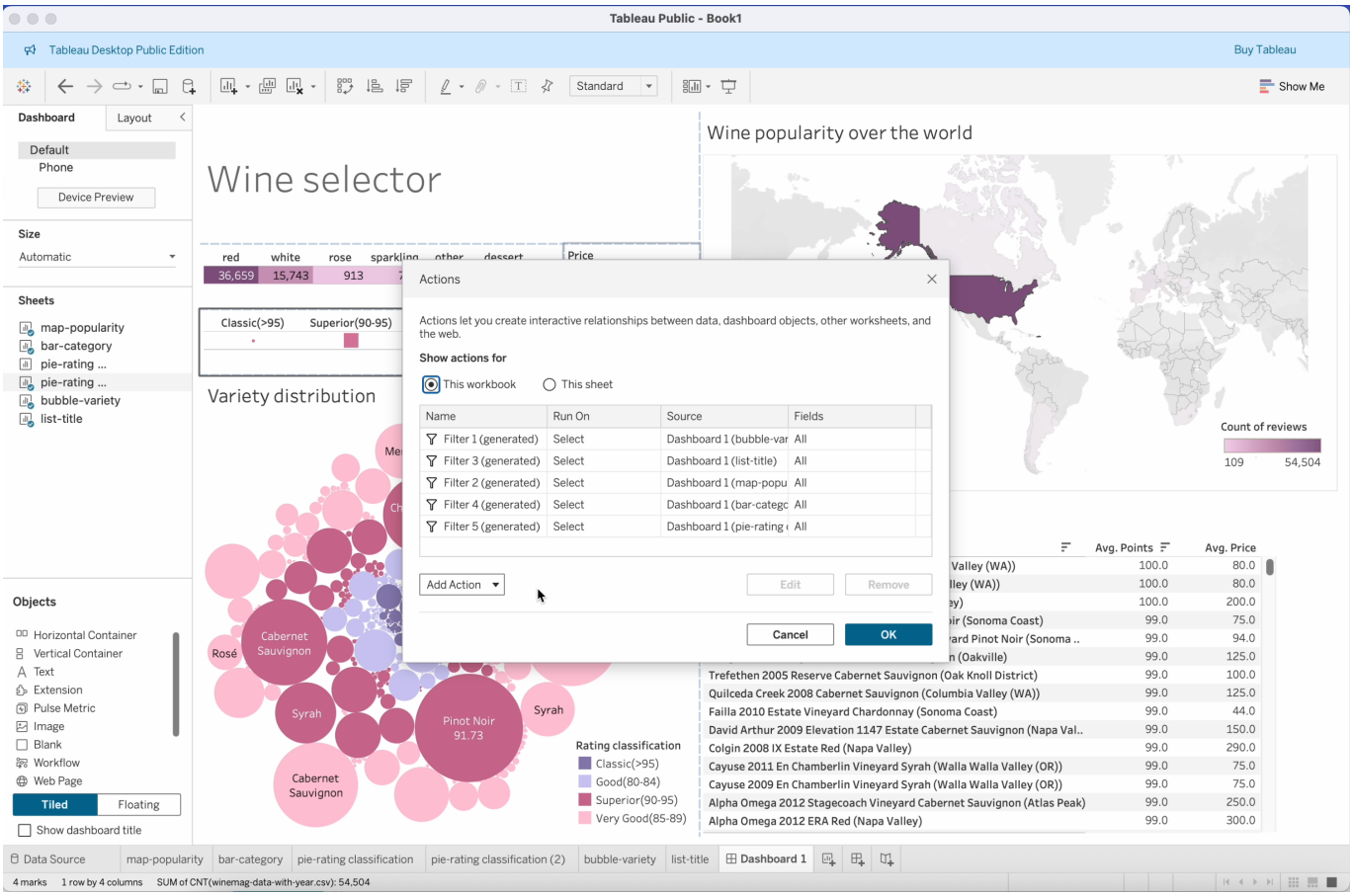
Example:

- Create a bar chart for **Rating Classification**
- Edit the **aliases** so labels are user-friendly (e.g., 95–100 → “Classic”)



3.3 Add Actions (Highlight, External Links and More)

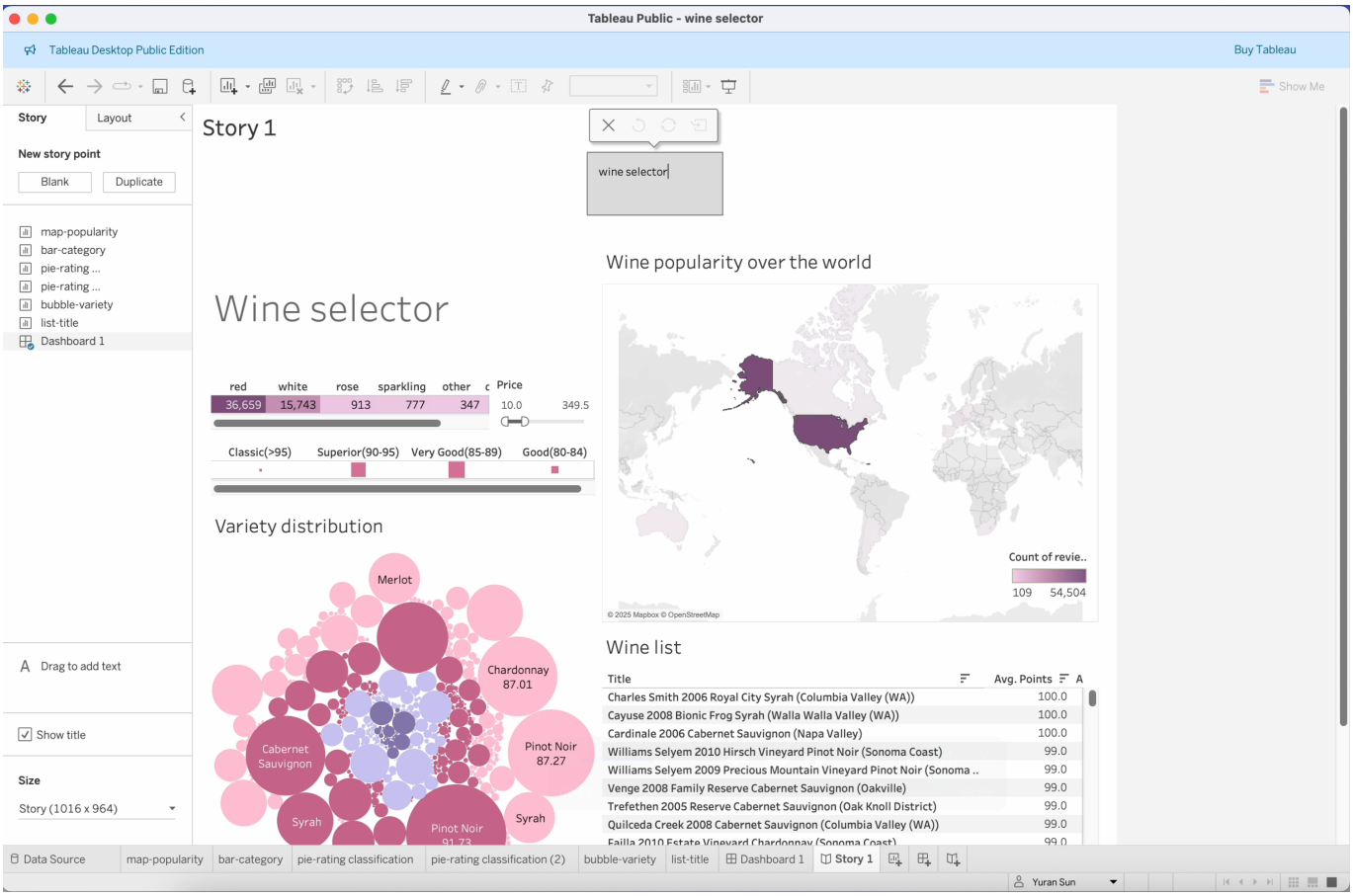
When we previously apply filters across sheets, actions are automatically added. Go to **Dashboard** → **Actions** and explore them.



Explore more actions:

- Highlight Action: Draw attention to specific data points without removing other data. Usually used when comparing highlighted data against the rest. (e.g., highlight a trend line from a country)
- Go to URL Action: open a Wikipedia page about a country or variety

3.4 (Optional) Turn into a Story



If you have created multiple dashboards or views, consider building a **Story** in Tableau. To do this, simply drag your dashboards and charts into a Story sheet. A Story lets you guide users step-by-step through your analysis, presenting insights in a clear, narrative format. This approach helps users follow a logical progression and better understand your findings.

4. Final Touches

- Arrange charts in a logical, easy-to-navigate layout.
- Rename all worksheet tabs with clear, descriptive names for clarity
- Use consistent colors, fonts, and formatting for a professional look.
- Add **captions** or **annotations** where needed
- Resize dashboard to **automatic** or standard sizes (e.g., 1200×800)
- Preview dashboard on **Tableau Public** to ensure all filters and interactions work smoothly.

Your Task: Build a US Winery Selector Dashboard

Imagine you are planning a trip to the United States and want to find a winery to visit. Your goal is to design an interactive dashboard in Tableau that helps users explore and select US wineries based on wine category, popularity, quality, and price. You will practice the techniques covered in class and may reuse charts from Lab 1's task.

Important: For all visualizations in this lab task, filter the dataset to include **only wines from the United States**. In each chart, set a filter on the `country` field and keep only **US** entries.

Please complete the following:

1. **Category Distribution Highlighted Table (Category Selector)**
   - Create a bar chart showing the distribution of wine categories (red, white, sparkling, etc.) in US.
   - Use this chart as a category selector in your dashboard.
2. **US State Wine Popularity Map (State Selector)**
   - Build a map showing wine review counts for each US state (`province`).
   - Use color to indicate popularity (number of reviews).
   - Use this map as a state selector.
3. **Price Slider Filter (Price Selector)**
   - Add a price slider filter to your dashboard.
   - Allow users to select wines within their preferred price range.
4. **Winery Bar Chart (Final Candidates Display)**
   - Create a bar chart showing US wineries and their average rating (`points`).
   - Use color to represent winery popularity (count of reviews).
   - Enable sorting by quality (average rating) or popularity (review count).
5. **Interactivity**
   - Make sure your dashboard allows users to filter and explore by category, state, and price.
   - Charts should update dynamically based on user selections.
6. **Naming and Organization**
   - Rename each charts clearly and descriptively.
   - Use clean formatting: readable labels, consistent colors, and logical sorting.
7. **Submission**
   - Save your dashboard to Tableau Public.
   - Name your workbook: **“US Winery Selector”**.
   - Submit your Tableau Public link on Moodle.